Deployed Binary-Size Cost Calculation

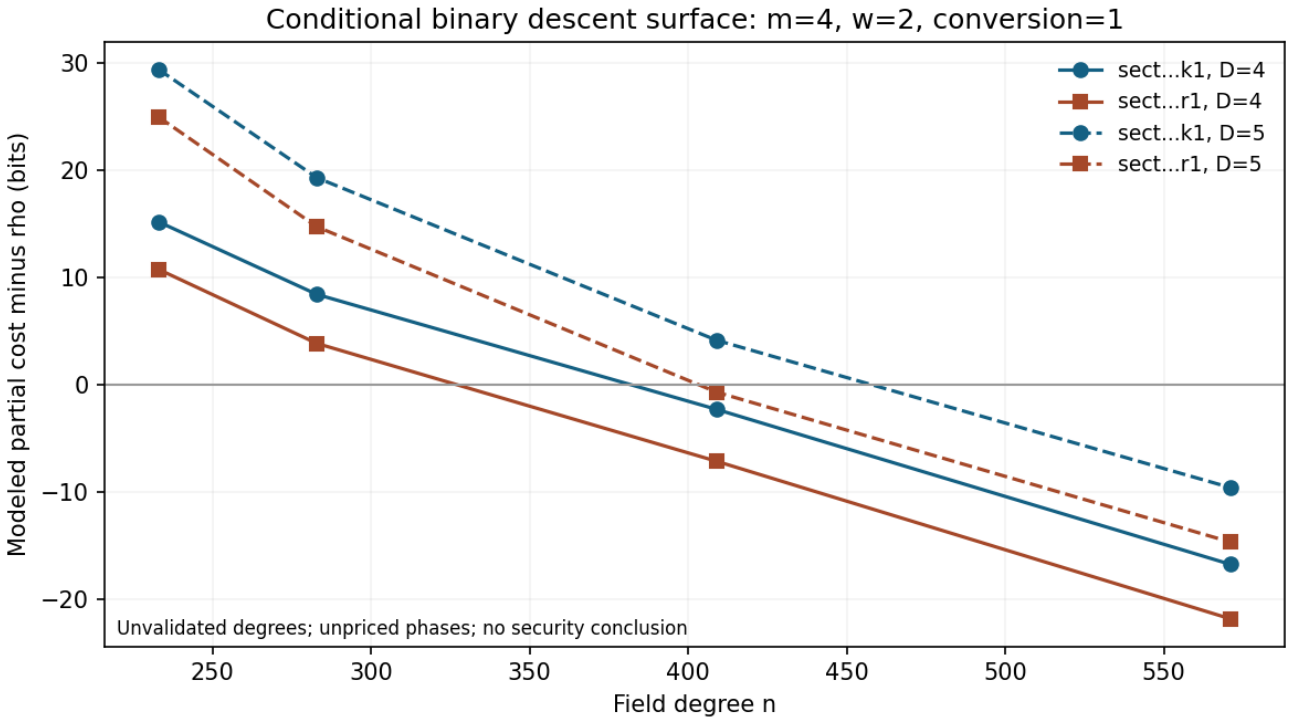
Research execution record

18 October 2026

Partial conditional model. No timing, attack or security claim.

The native calculator produced 1,944 rows under the original frozen protocol, and 2,592 under an additive amendment including zero-charge sensitivity. Nine curve cases, four arities, six degree models, three column-work exponents and four conversion weights occur in the amended output. Four native checks pass, including reproduction of the historical $n = 131$ anchors.

For $m = 4$, $w = 2$, conversion one and constant degree four, the partial-cost gaps on the Koblitz curves of degrees 233, 283, 409 and 571 are respectively $+15.20, +8.42, -2.30, -16.75$ bits against idealized automorphism rho. The model includes unsuccessful attempts and relation algebra and searches all integer subspace dimensions $4 \leq \ell \leq n$.



Degree four at deployed sizes is unvalidated. Conversion from column work to group operations is assumed, and factor-base construction, successful extraction and other overhead are absent. Degree five changes the crossover. Coordinate-storage estimates are not measured peak memory. Neither a modeled negative gap nor a zero-charge arm establishes a usable attack.

Reproduction and evidence. The standalone Rust crate, frozen protocol, additive amendment, original and amended JSON, source hashes and exact commands are retained beside this report. The original model executed at revision `04c3c012`; the amendment at `b3a3321c`. The initial missing-binary launch exited 127 and is recorded separately. Python renders stored JSON only.